

Revised Load Interconnection System Impact Study Report

For Queue Project Q23-2328

North Adamsville 138 kV
May 29, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2328. The Interconnection Customer proposes a 304.2 MW large load (data center) facility interconnecting at the North Adamsville 138 kV substation in the Kentucky area. The request was received on June 28, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 418 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$72,279,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BANGHAM / JUDSON 138kV	ST. HELENS / NORTH ADAMSVILLE 138kV Line	189.5	172	100.0
ST. HELENS / N ADAMSVILLE SUBSTATION 138kV line	PEACH VLY TIE / PINN RD 138kV Line	186.0	197	100.0
THERMOPOLIS / JUDSON 138kV	N ADAMSVILLE / ST. HELENS 138kV	179.7	184	100.0
BANGHAM / NORTH ADAMSVILLE 138kV	N ADAMSVILLE / ST. HELENS 138kV	176.8	180	100.0
MAINWARING / ST. HELENS 138kV	OJO / PORT MADISON 138kV	176.4	241	69.5
THERMOPOLIS / TUSCOLA E 138kV	N ADAMSVILLE / ST. HELENS 138kV	171.4	200	100.0
PEACH VALLEY TIE / PINN RD 138kV	N ADAMSVILLE / ST. HELENS 138kV line	169.5	216	100.0
PEACH VALLEY TIE / TUSCOLA EAST 138kV	N ADAMSVILLE / ST. HELENS 138kV line	167.5	213	100.0
PINN ROAD / PORT MADISON 138kV L	SNOHOMIH / SAND SPRINGS 138kV line	127.4	333	38.1
PORT MADISON / OJO 138kV	SAND SPRINGS / SNOHOMISH 138kV	126.9	342	38.1
SNOHOMISH / MAINWARING 138kV	PORT MADISON / OJO 138kV	114.0	489	46.2
SNOHOMISH / SAND SPRINGS 138kV	PORT MADISON SUBSTATION / OJO 138kV	113.4	500	46.2
SAND SPRINGS / RUSSELLVILLE E 138kV Line	NORTHFIELD / BERRY STREET 345kV	105.8	357	24.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BANGHAM / JUDSON 138kV	Rebuild line with higher-capacity conductor	\$5,745,000	100% (\$5,745,000)
ST. HELENS / N ADAMSVILLE SUBSTATION 138kV line	Rebuild line with higher-capacity conductor	\$3,238,000	100% (\$3,238,000)
THERMOPOLIS / JUDSON 138kV	Rebuild line with higher-capacity conductor	\$6,614,000	100% (\$6,614,000)
BANGHAM / N ADAMSVILLE 138kV	Rebuild line with higher-capacity conductor	\$8,562,000	100% (\$8,562,000)
ST. HELENS / MAINWARING 138kV	Rebuild line with higher-capacity conductor	\$13,064,000	80% (\$10,422,000)
THERMOPOLS / TUSCOLA E 138kV	Rebuild line with higher-capacity conductor	\$6,231,000	100% (\$6,231,000)
PEACH VLY TIE / PINN RD 138kV	Rebuild line with higher-capacity conductor	\$3,401,000	100% (\$3,401,000)
PEACH VALLEY TIE / TUSCOLA EAST 138kV	Rebuild line with higher-capacity conductor	\$7,437,000	100% (\$7,437,000)
PINN ROAD / PORT MADISON 138kV L	Rebuild line with higher-capacity conductor	\$3,557,000	54% (\$1,905,000)
PORT MADISON / OJO 138kV L	Rebuild line with higher-capacity conductor	\$17,005,000	54% (\$9,108,000)
SNOHOMISH / MAINWARING 138kV	Rebuild line with higher-capacity conductor	\$9,548,000	35% (\$3,314,000)
SNOHOMISH / SAND SPRINGS 138kV	Rebuild line with higher-capacity conductor	\$15,734,000	35% (\$5,461,000)
SAND SPRINGS / RUSSELLVILLE E 138kV Line	Reconductor line	\$4,902,000	17% (\$841,000)
	Total allocated to project		\$72,279,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.